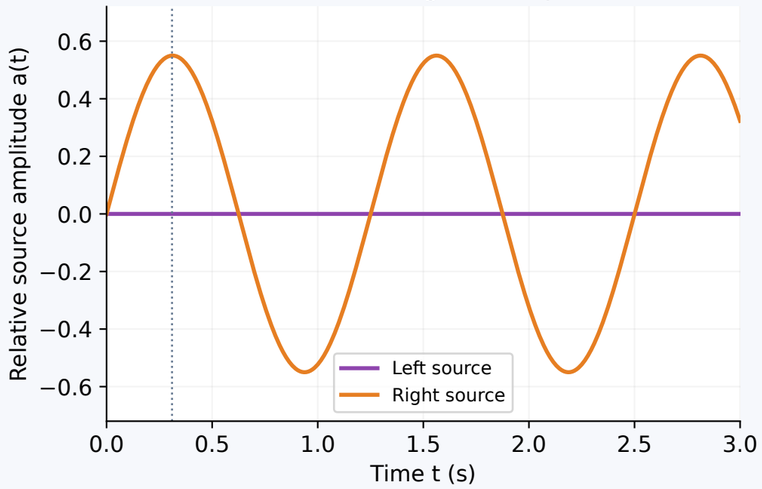
MAGNETIC INPUT → FLUID FLOW → PARTICLE RESPONSE

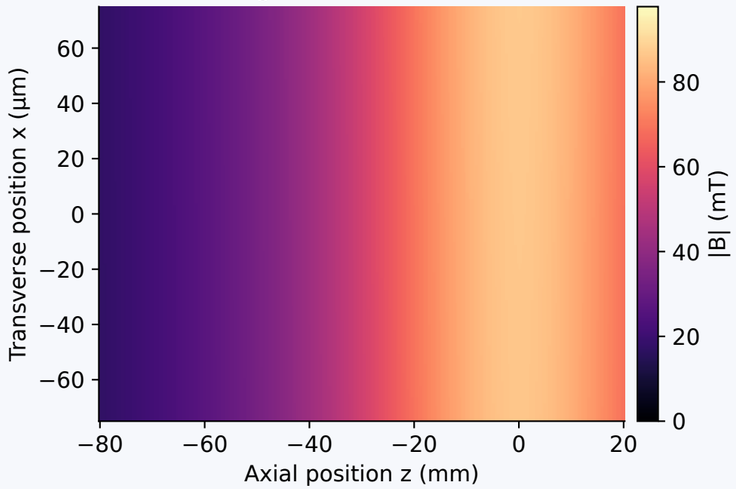
# Sinusoidal right source

$a_L = 0, \quad a_R = 0.55\sin(2\pi\,0.8t)$

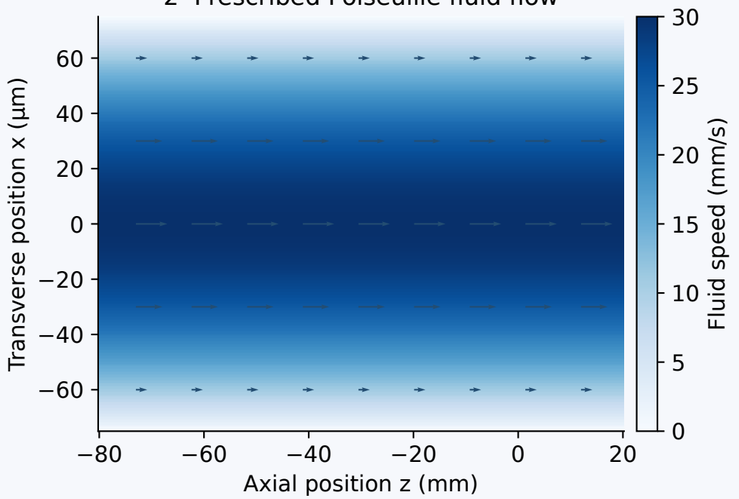
1a Prescribed magnetic amplitudes



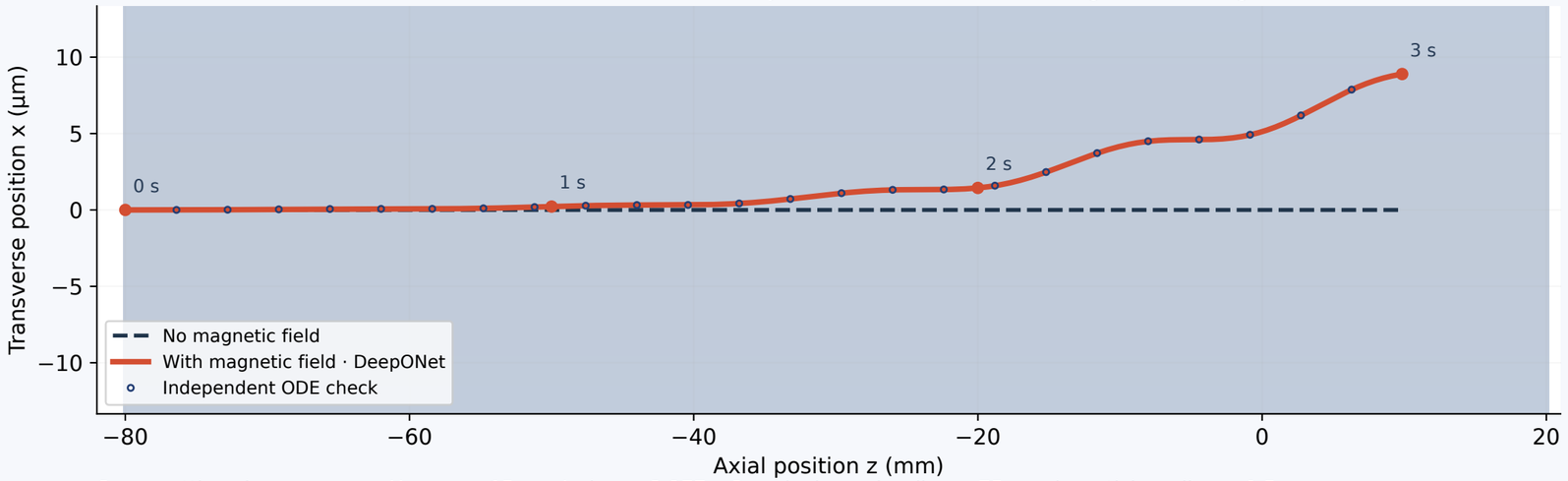

1b Magnetic field at t = 0.31 s



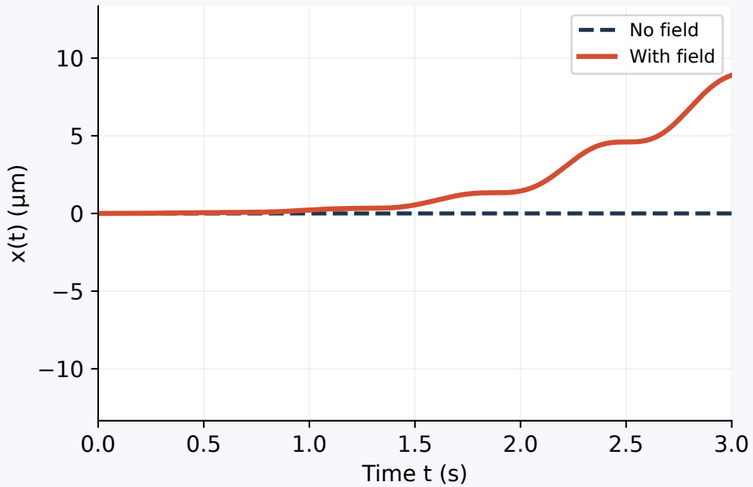

2 Prescribed Poiseuille fluid flow




3a Same release and flow: trajectories with / without magnetic forcing




3b Transverse particle displacement




Same settings in every case: Umean = 15 mm/s | η = 3.277 mPa·s | channel radius = 75 μm | particle radius = 0.5 μm

Time-varying amplitudes; fixed spatial source geometry. Overdamped induced-dipole model. Axial and transverse axes use different units.

Independent max error: x 0.008 μm · z 0.57 μm